

Project Target Region-India

My project focuses on India, particularly peri-urban and rural communities, due to several critical factors:

Urgent Sanitation Needs

Many regions in India face significant challenges in accessing clean water and adequate sanitation infrastructure, leading to pollution and health risks from untreated domestic wastewater.

Demand for Decentralized Solutions

Traditional centralized wastewater treatment systems are often financially and technically unfeasible for small or rural communities. A low-cost, community-scale bioreactor provides a practical, decentralized alternative.

Alignment with Sustainable Development Goals

This initiative directly supports UN Sustainable Development Goal 6 (Clean Water and Sanitation), promoting equitable access to safe water and sanitation in underserved communities.

Favorable Climatic Conditions

India's generally high solar irradiation and warm temperatures provide ideal conditions for algal growth, improving the overall performance and productivity of the bioreactor.

Empowerment and Resource Recovery

Implementing a community-based photobioreactor can foster local self-reliance in wastewater management, while also offering opportunities for resource recovery through the reuse of harvested algal biomass.

Bioreactor Panel Materials Selection

Introduction

For the transparent bioreactor panels, **Acrylic (PMMA)** and **Polycarbonate (PC)** have been identified as primary candidate materials. Both materials are transparent, lightweight, and importantly, relatively low cost, making them suitable for applications in resource-limited settings.

Comparison Table

Criterion	Acrylic (PMMA)	Polycarbonate (PC) - Multiwall	Notes/Consideration (Pros and Cons)
Cost	Very Good (approx. \$8.50/m ² based on local retail in India.	Good (significantly cheaper - \$12/m ²)	- Varying cost, Acrylic considered cheaper than Polycarbonate
Light Transmission	Good clarity	Lower clarity and potential cleaning challenges	- Suggests with lower price, lower quality but could still be used if recycled or changed often
Chemical Resistance	It generally has good chemical resistance, but it can be sensitive to some alcohol and solvents. It holds up well to water and the typical nutrients used in bioreactors.	This material is also generally resistant to many chemicals. However, it can be affected by strong bases or certain solvents. Like acrylic, it's typically good for use with water and nutrients in bioreactor systems.	- Both have a strong resistivity to chemicals and are well suited in high nitrogen and phosphorus environments.
Structural Strength / Impact Resistance	It has 17 times more impact resistivity than glass, can resist breakage through shock and impact.	It has 250 times more impact resistivity than glass.	- Both have a strong structural resistivity with polycarbonate having a stronger resistivity than acrylic
UV Stability / Resistance	Very good UV stability, but over long periods of	Moderate (Requires UV coating for long-	- Both require coating with long term UV

	continuous outdoor exposure, it could degrade. But with additives and coating, this could be avoided.	term outdoor use to prevent yellowing)	light exposure to prevent yellowing but overall Acrylic is good at handling long UV exposure
Ease of Fabrication	Easy to cut (standard tools), polishes well, bonds easily with solvent cement.	It is harder to cut, drill cleanly, and bond transparently without specialized methods.	- Acrylic could be a good, standardized material as it is easy to set up
Sustainability	Recyclable, though recycling streams/infrastructure vary.	Recyclable, though recycling streams/infrastructure vary.	- Both are considered recyclable and can safely be used without harming the environment.
Non-Toxicity to Algae	Excellent; widely used in aquariums and lab settings; inert.	Excellent; widely used in aquariums and lab settings; inert	- Both are excellent in treating algae medium

Justification

This section provides a formal rationale for selecting the transparent panel material in a modular algae-based photobioreactor, with a focus on balancing initial cost, long-term durability, and functional performance for a community-scale application in India.

1. Primary Material Recommendation

Multiwall polycarbonate (PC) is recommended as the preferred material for transparent bioreactor panels.

2. Key Justifications

Durability and Lifecycle Cost-Effectiveness

Although multiwall PC presents a higher initial material cost (approximately \$12/m²) compared to acrylic (\$8.50/m²), its superior impact resistance significantly reduces the likelihood of damage and the associated maintenance or replacement costs. This advantage supports long-term economic sustainability for community-managed systems.

Structural Resilience

Multiwall PC demonstrates impact resistance up to 250 times greater than glass, providing substantially higher robustness in outdoor and community settings than acrylic (which is roughly 17 times stronger than glass but more prone to cracking).

Sufficient Light Transmission

Multiwall PC achieves adequate light penetration to sustain algae growth, even though it does not offer the same crystal clarity as solid polycarbonate sheets. Its proven use in greenhouse construction within India supports its suitability for light-dependent photobioreactor operations.

Practicality and Local Availability

The structural rigidity, moderate weight, and established supply chain of multiwall PC in the Indian market make it a practical choice for community-level fabrication, assembly, and maintenance using basic tools and skills.

3. Alternatives and Trade-Offs

Acrylic (PMMA) was evaluated for its low initial cost and ease of fabrication. However, its relatively low impact resistance increases the risk of breakage and recurrent replacement, which may compromise long-term cost-efficiency and system reliability. Solid polycarbonate, while offering superior clarity and impact resistance, is prohibitively expensive for widespread use in community-based systems and often presents logistical sourcing challenges. Multiwall PC represents an acceptable compromise on optical clarity while delivering significant benefits in durability and cost over the system's lifetime.

4. Alignment with Project Goals

Multiwall polycarbonate achieves a practical balance between cost-effectiveness, durability, and adequate light transmission, making it the most appropriate material for a replicable, sustainable, and resilient community-driven wastewater treatment photobioreactor in the Indian context.